



Inspiring Young Minds
Through Knowledge Olympiads

CLASS
6

SET-A

DO NOT OPEN THIS BOOKLET UNTIL ASKED TO DO SO

Name:.....

Section:..... Roll No.:.....

Contact No.....

Total Questions: 50

Time: 1 hr.



10TH SOF INTERNATIONAL MATHEMATICS OLYMPIAD

Guidelines for the Candidate

1. You will get additional ten minutes to fill up information about yourself on the OMR Sheet, before the start of the exam.
2. Write your **Name, School Code, Class, Section, Roll No. and % of marks/grade** in last class clearly on the **OMR Sheet** and do not forget to sign it.
3. The Question Paper comprises four sections :

Logical Reasoning (15 Questions), **Mathematical Reasoning** (20 Questions), **Everyday Mathematics** (10 Questions) and **Achievers Section** (5 Questions)

Each question in Achievers Section carries 3 marks, whereas all other questions carry one mark each.

4. All questions are compulsory. There is no negative marking. Use of calculator is not permitted.
5. There is only ONE correct answer. Choose only ONE option for an answer.
6. To mark your choice of answers by darkening the circles in the OMR Sheet, use **HB Pencil** or **Blue / Black ball point pen** only. E.g.

Q. 16: Rahul bought 4 kg 90 g of apples, 2 kg 60 g of grapes and 5 kg 300 g of mangoes. The total weight of all the fruits he bought is _____.

A. 11.450 kg

B. 11.000 kg

C. 11.850 kg

D. 11.250 kg

As the correct answer is option A, you must darken the circle corresponding to option A in the OMR Sheet.

16. ● (A) (B) (C) (D)

7. Rough work should be done in the blank space provided in the booklet.
8. Return the OMR Sheet to the invigilator at the end of the exam.
9. Please fill in your personal details in space on top of this page before attempting the paper.

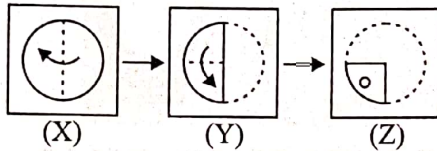


1. Find the missing number, if same rule is followed in all the three figures.

$25 : 5$ $36 : 6$ $? : 7$

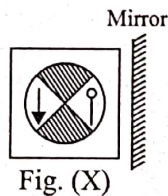
- A. 49
B. 37
C. 50
D. 94

2. A set of three figures X, Y and Z showing a sequence of folding a piece of paper is given. Fig. (Z) shows the manner in which the folded paper has been cut. Select the option which shows the unfolded form of Fig. (Z).



- A.
B.
C.
D.

3. Select the correct mirror image of Fig. (X).



- A.
B.
C.
D.

4. Select odd one out.

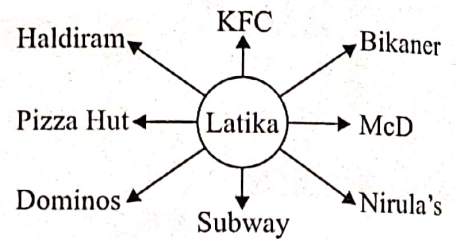
- A. Lion
B. Cat
C. Rabbit
D. Fox

5. Which of the following net can be used to form the given cube?



- A.
B.
C.
D.

6. Latika is facing Bikaner. What will she be facing, if she turns 315° anti-clockwise?



- A. McD
B. KFC
C. Subway
D. Nirula's

7. If 'earth' is called 'sky', 'sky' is called 'tree' and 'tree' is called 'wall', then on which of the following a fruit grows?

- A. earth
B. sky
C. wall
D. tree

8. In which of the following options, Fig. (X) is exactly embedded as one of its part?

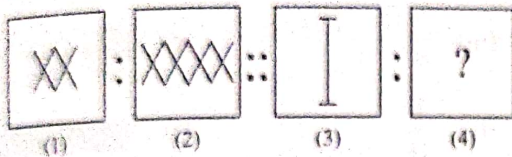


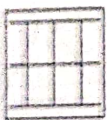
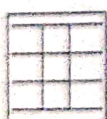
Fig. (X)

- A.

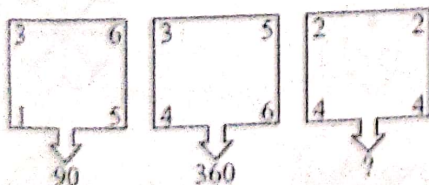
- B. 
- C. 
- D. 

9. There is a certain relationship between figures (1) and (2). Establish the same relationship between figures (3) and (4) by selecting a suitable figure from the options which will replace the (?) in figure (4).



- A. 
- B. 
- C. 
- D. 

10. Find the missing number.



- A. 64
- B. 400
- C. 360
- D. 380

11. How many circles are there in the figure given below?



- A. 9
- B. 10
- C. 11
- D. 12

12. Which of the following options will complete the Fig. (X)?

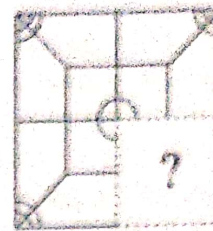
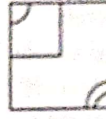
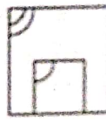


Fig. (X)

- A. 
- B. 
- C. 
- D. 

13. Naman ranks 14th from the top and 26th from the bottom in a class. How many students are there in the class?

- A. 31
- B. 39
- C. 34
- D. 33

14. If $8 * 7 = 56$, $9 * 6 = 54$ and $8 * 3 = 24$, then find the value of $8 * 5$.

- A. 55
- B. 40
- C. 50
- D. 44

15. One morning Raju start walk towards his school. After covering 5 km distance in East, he turned to the left and walk 2 km, then again turns to the right and walk 5 km. He again turns to the left. In which direction is he facing now?

- A. East
- B. West
- C. North
- D. South

16. The simplification of $2.002 + 7.9 \{2.8 - 6.3(3.6 - 1.5) + 15.6\}$ yields
- 2.002
 - 4.2845
 - 40.843
 - 42.845

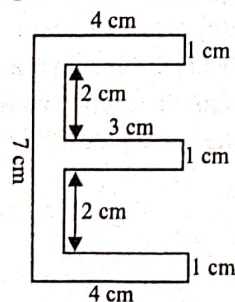
17. If $I = \frac{13}{4} \div \frac{5}{6}$,
 $II = 13 \div [(4 \div 5) \div 6]$,
 $III = [13 \div (4 \div 5)] \div 6$ and
 $IV = 13 \div 4(5 \div 6)$ then, which of the following are equal?
- I and II
 - I and IV
 - I and III
 - All are equal

18. Which of the following statements is/are INCORRECT?
- Eighty four in Roman numeral is written as CXXXIV.
 - There are seven zeroes in 1 crore.
 - There are ten thousand milligrams in 1 kg.
 - The smallest 4-digit number formed by using all the digits 4, 3, 0, 8 without repetition is 3048.
- Both (i) and (iii)
 - Both (i) and (iv)
 - Only (i)
 - Only (iii)

19. Using distributive property, $258 \times 1008 = ?$
- $258 + 1000 + 8$
 - $258 \times 1000 + 258 \times 8$
 - $258 \times 1000 + 8$
 - $1000 + 8 \times 258$

20. Which situation is best represented by the integer -12 ?
- The length of curtains is increased by 12 inches.
 - The price of petrol is reduced by ₹ 12.
 - The size of picture is enlarged by 12 per cent.
 - The temperature rises by 12°C .

21. Find the area of the following figure (not drawn to scale) by splitting them into rectangles and squares.



- 14 sq. cm
- 15 sq. cm
- 17 sq. cm
- 16 sq. cm

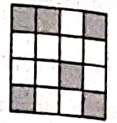
22. Match the fractions given in Column I with the shaded portion of figures in Column II.

Column I

Column II

(i) $\frac{6}{4}$

(p)



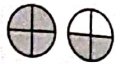
(ii) $\frac{6}{10}$

(q)



(iii) $\frac{6}{6}$

(r)



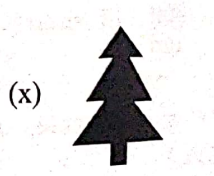
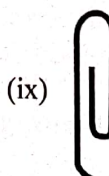
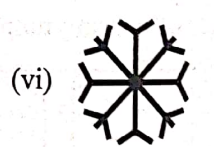
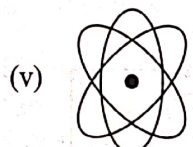
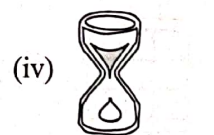
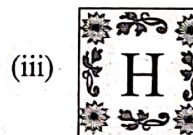
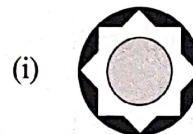
(iv) $\frac{6}{16}$

(s)



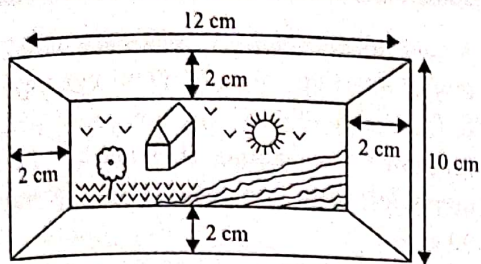
- | | | | |
|--------|------|-------|------|
| (i) | (ii) | (iii) | (iv) |
| A. (r) | (q) | (s) | (p) |
| B. (s) | (p) | (r) | (q) |
| C. (r) | (s) | (p) | (q) |
| D. (q) | (p) | (s) | (r) |

23. How many of the following figures are symmetrical?



- 9
- 10
- 3
- None of these

24. The diagram below shows a framed picture. Find the area of the picture.



- A. 48 cm^2
 B. 39 cm^2
 C. 92 cm^2
 D. 120 cm^2

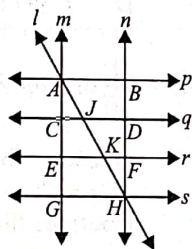
25. Which sign will show correct comparison between the given expressions?

$$(-3) - 74 + (-42) + (-82) \quad \square \quad (-12) + (-43) - (-57) + 1.$$

- A. $>$
 B. $<$
 C. $=$
 D. Can't be determined

26. The number of collinear points in the given figure are

- A. 10
 B. 18
 C. 15
 D. 12



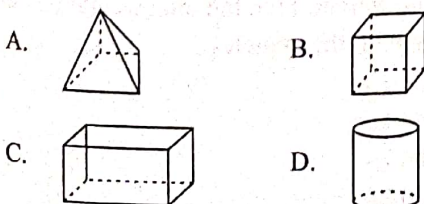
27. A polygon has prime number of sides. If number of sides is equal to the sum of the two least consecutive primes. The number of diagonals of the polygon is

- A. 4
 B. 5
 C. 7
 D. 10

28. Number of primes between 16 to 80 and 90 to 100 is

- A. 20
 B. 18
 C. 17
 D. 16

29. Which three-dimensional figure has 5 faces, 8 edges and 5 vertices?



30. If the sum of two angles is equal to an obtuse angle, then which of the following is not possible?

- A. One obtuse and one acute angle
 B. One right angle and one acute angle
 C. Two acute angles
 D. Two right angles

31. Latika, Garima and Sanchi draw 3 cards each from 9 cards numbered from 1 to 9.

Latika : L, Garima : G, Sanchi : S

L : The product of my numbers is 48.

G : The sum of my numbers is 15.

S : The product of my numbers is 63.

What is the largest number in the cards of Sanchi?

- A. 8
 B. 7
 C. 9
 D. 6

32. Find the greatest number from the options that will divide 1025, 1299 and 1575 leaving remainders 5, 7 and 11 respectively.

- A. 70
 B. 78
 C. 68
 D. 98

33. Select the CORRECT statement.

- A. 1592 rounded off to nearest tens is 1600.
 B. 7532 rounded off to nearest thousands is 7500.
 C. 245 rounded off to nearest hundreds is 240.
 D. 214653 rounded off to nearest ten thousands is 210000.

34. In an activity class, students were asked to make a circular rangoli. Trishu, Sidak, Mini and Kavleen made

rangolies having different diameters as $\frac{17}{20}$ inches, $\frac{3}{4}$ inches, $\frac{5}{6}$ inches and $\frac{7}{10}$ inches, respectively.

Who among them made the smallest one?

- A. Kavleen
 B. Trishu
 C. Mini
 D. Sidak

35. Which of the following will not represent zero?

- A. $\frac{1}{0}$
 B. 0×9
 C. $\frac{0}{2}$
 D. $\frac{3-3}{2}$

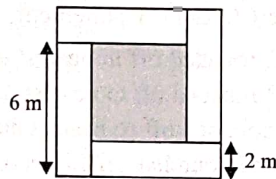
36. Mohit travelled $\frac{2}{3}$ of his journey by train, $\frac{1}{4}$ of the journey by bus and the remaining journey by car. If the distance he travelled by train is 60 km more than the distance he travelled by bus, what was the total distance he had travelled?

A. 70 km
B. 250 km
C. 144 km
D. 153 km

37. Deepika has more than 30 stickers but less than 40 stickers. She can pack the stickers into packs of 2, 3 or 4 without leaving any remainder. How many stickers does she have?

A. 36
B. 37
C. 33
D. 39

38. Priya used four similar rectangular carpets and a square carpet for the flooring in her study room as shown below.



If it cost her ₹ 6 per square metre for the rectangular carpets and ₹ 8 per square metre for the square carpet, how much did she pay in all?

A. ₹ 512
B. ₹ 600
C. ₹ 417
D. ₹ 416

39. Kiara weighs 34.5 kg. Her uncle weighs thrice as much as Kiara. What is their total weight?

A. 69 kg
B. 103.50 kg
C. 138 kg
D. 172.50 kg

40. Two ropes 16 m and 20 m long are to be cut into small pieces of equal lengths. What will be the maximum length of each piece?

A. 5 m
B. 4 m
C. 7 m
D. 10 m

41. A cold storage company had a stock of 58970252 egg trays. Out of this stock, 3789441 egg trays were sent to Delhi and 4207985 egg trays were sent to Punjab for public consumption. How much is the exact egg count left with company, if an egg tray contains 30 eggs?

A. 1529184780
B. 1509184780
C. 1529084780
D. 1529184880

42. Garima was shopping for a new hand bag. The one she wanted to buy cost ₹ 428.98. The sales person informed her that the same hand bag would be on sale the following week for ₹ 399.99. How much money would she save by waiting until the hand bag went on sale?

A. ₹ 28.99
B. ₹ 29.01
C. ₹ 128.99
D. ₹ 39.09

43. $\frac{1}{10}$ of a rod is coloured red, $\frac{1}{20}$ orange, $\frac{1}{30}$ yellow, $\frac{1}{40}$ green, $\frac{1}{50}$ blue, $\frac{1}{60}$ black and the rest violet. If the

length of the violet portion is 12.08 m, then what is the length of the rod?

A. 16 m
B. 18 m
C. 20 m
D. 30 m

44. A drum of oil has a capacity of 705 litres. Two containers of 135.75 litres and 253.50 litres capacity are filled with oil from the drum. Find the quantity of oil left in the drum.

A. 720 litres
B. 300 litres
C. 215.75 litres
D. 315.75 litres

45. A truck can carry 582 boxes of biscuits weighing 16 kg each, whereas a van can carry 359 boxes each of the same weight. Find the total weight that can be carried by both the vehicles.

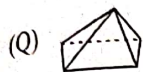
A. 20000 kg
B. 15056 kg
C. 25000 kg
D. 12000 kg

ACHIEVERS SECTION

46. Match the following.



(i) Octagonal Prism



(ii) Triangular Prism



(iii) Rectangular Pyramid



(iv) Pentagonal Pyramid

- A. (P) – (i), (Q) – (ii), (R) – (iv), (S) – (iii)
 B. (P) – (ii), (Q) – (iii), (R) – (iv), (S) – (i)
 C. (P) – (ii), (Q) – (iv), (R) – (iii), (S) – (i)
 D. (P) – (ii), (Q) – (iii), (R) – (i), (S) – (iv)

47. Trishika bought some strawberries. $\frac{1}{8}$ of the strawberries were rotten and had to be thrown away. Trishika then used 25 strawberries to bake a cake.

She then had $\frac{1}{4}$ of the total strawberries left.

- (a) How many strawberries did she have at first?
 (b) If Trishika shared the remaining strawberries between her two sons equally, what fraction of the strawberries did each of them receive?

- (a) (b)
 A. 70 $\frac{1}{8}$
 B. 50 $\frac{4}{8}$
 C. 40 $\frac{1}{8}$
 D. 40 $\frac{1}{4}$

48. Consider the following statements :

- I. Every prime is odd.

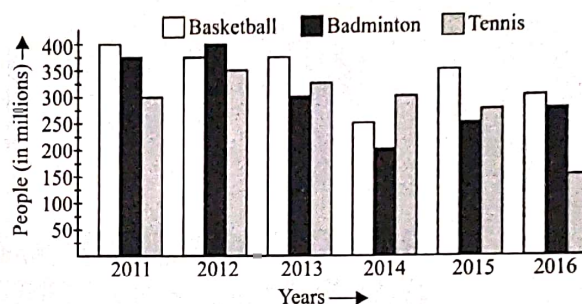
II. Product of any two prime numbers is odd.

Which of the given statement(s) is/are correct?

- A. I alone
 B. II alone
 C. I and II
 D. Neither I nor II

Direction (49-50) : Study the bar graph given below and answer the questions.

(Preferences of people in playing different games over the years.)



49. From 2011 to 2016, find the difference between the total number of people who preferred to play Basketball and Badminton (in millions)?

- A. 525
 B. 275
 C. 730
 D. 250

50. How many people (in millions) have preferred to play Tennis in all the years altogether?

- A. 3200
 B. 2100
 C. 1200
 D. 1700

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